Cascaded $\chi^{(2)}{:}\chi^{(2)} \to \chi^{(3)}_{\text{eff}}$

in a Doubly-Resonant Dielectric Metasurface

2026-05-14

Outline

1. **The cascade idea** — why and what
2. **Setup** — doubly-resonant metasurface
3. **Theory** — TCMT, adiabatic elimination, bandwidth sum rule
4. **Comparison with native $\chi^{(3)}$ resonance** — SPM, bistability, FWM, XPM, sign tunability
5. **Cascade-only capabilities** — saturable absorber, AM modulator, optical limiter, sensor
6. **Feasibility** — host materials, bulk vs metasurface
7. **Design knobs** — what the metasurface group needs to deliver
8. **Summary and asks**

1. The cascade idea

What we want, and the obstacle

Goal. A large effective Kerr response $\chi^{(3)}_{\text{eff}}$ on the **fundamental frequency** ω_1, driven by a single CW pump at ω_1.

Obstacle. Low-loss dielectrics with good $\chi^{(2)}$ (LiNbO$_3$, GaP) have a *small* native $\chi^{(3)} \sim 10^{-21}\,\mathrm{m}^2/\mathrm{V}^2$.

Standard fix: push to a high-$\chi^{(3)}$ host (Si, AlGaAs near band edge), but those bring thermal / free-carrier / TPA penalties.

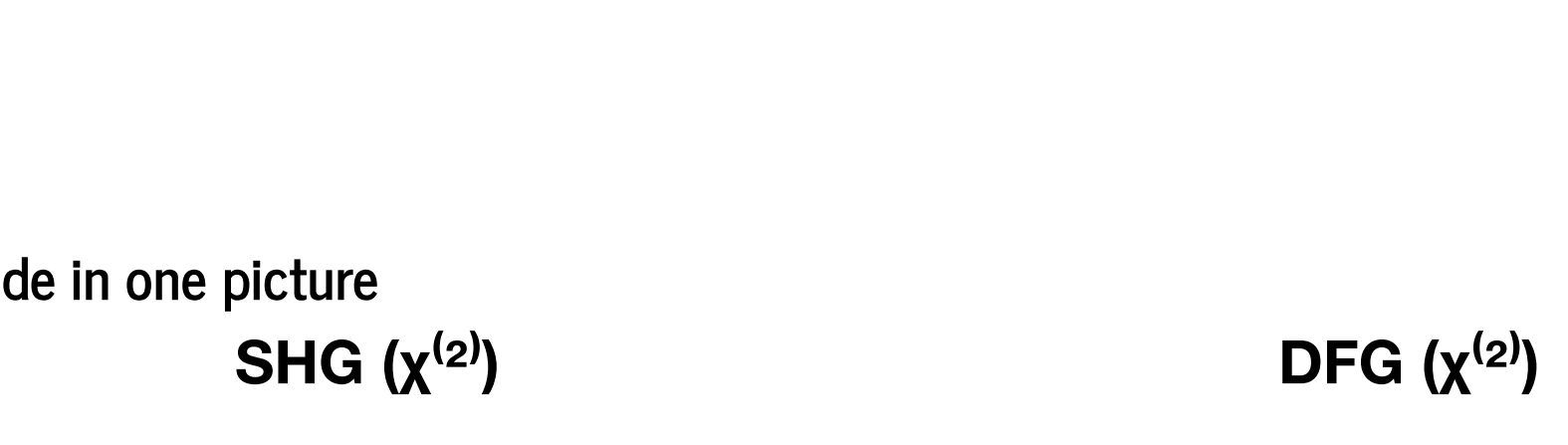
Cascade alternative. Use $\chi^{(2)}$ *twice*:

$$\omega_1 + \omega_1 \xrightarrow{\chi^{(2)}} (2\omega_1) \xrightarrow{\chi^{(2)}} \omega_1 + \omega_1$$

The SH at $2\omega_1$ is *virtual* (off-resonant from a real SH cavity); the round-trip back-action on the FM looks like a Kerr term.

Pay-off. Order-of-magnitude $\chi^{(3)}_{\text{eff}}$ enhancement on the FM, plus several knobs (sign-tunability, AM modulation, sensing) that native Kerr cannot offer.

The cascade in one picture



Net: $\omega_1 + \omega_1 \to (2\omega_1) \to \omega_1 + \omega_1$ — effective $\chi^{(3)}$ on the fundamental.

Energy-level diagram of the cascade. SHG (up arrows) virtually populates the SH level at $2\omega_1$ off-resonant from the BIC at ω_2 by $\Delta\omega = 2\omega_1 - \omega_2$; DFG (down arrows) returns to two ω_1 photons. The SH propagator $G(2\omega_1) \sim 1/(\Delta\omega + i\gamma_2/2)$ controls the strength of the cascade.

> 💡 **Tip**
>
> The SH is **never radiated** — it stays inside the high-Q BIC and back-reacts on the FM. This is the experimental signature: an intensity-dependent FM response without measurable SH light leaving the sample.

2. Setup

The doubly-resonant metasurface

Two co-localised resonances in one meta-atom array:

- **FM resonance** at ω_1: a *quasi-BIC* (qBIC) with controlled radiative coupling $\gamma_1^{(\text{rad})}$ — engineered for efficient in-coupling at the pump frequency.
- **SH resonance** at $\omega_2 \approx 2\omega_1$: a (quasi-)BIC with $\gamma_2 \simeq \gamma_2^{(\text{nr})}$ dominated by intrinsic absorption — maximum $Q_2 = \omega_2/\gamma_2$.

Decay-rate hierarchy (typical design):

$$\gamma_1^{(\text{rad})} \sim \gamma_1^{(\text{nr})} \ll \gamma_2 \ll \omega_1$$

Material requirements

- Non-centrosymmetric crystal ($\chi^{(2)} \neq 0$): LiNbO$_3$, GaP, AlGaAs, AlN
- Engineered to break inversion symmetry of the meta-atom *and* permit a non-zero modal overlap (see §7).

What we do NOT want

- SH out-coupling (a "good cascade implementation" produces **no measurable SH** at the output).
- The SH cavity is heat-sunk — TPA-like loss is dissipated in the SH mode and does *not* dump into the FM port.

Conventions

These are pinned once for the whole derivation:

- **Real fields** in Boyd's convention: $\mathcal{E}(t) = \frac{1}{2}[Ee^{-i\omega t} + \text{c.c.}]$.
- **Boyd degeneracy factors:**

$$P^{(2)}(2\omega_1) = \varepsilon_0\chi^{(2)}E^2 \quad (\text{SHG, no factor})$$
$$P^{(2)}(\omega_1) = 2\varepsilon_0\chi^{(2)}E(2\omega_1)E^*(\omega_1) \quad (\text{DFG, factor 2})$$
$$P^{(3)}(\omega_1) = 3\varepsilon_0\chi^{(3)}|E|^2E \quad (\text{SPM, factor 3})$$

- **Brillouin energy normalization** for modes (dispersion of ε_r matters):

$$\tfrac{1}{2}\int [\,\varepsilon_0\tilde{\varepsilon}_n|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2\,]\,d^3r = 1$$

  with $\tilde{\varepsilon}_n = \partial(\omega\varepsilon_r)/\partial\omega$ at ω_n.
- **Kleinman symmetry** for $\chi^{(2)}_{ijk}$ — one tensor for both SHG and DFG.

> ⚠ **Warning**
>
> Different effective refractive indices at ω_1 and $2\omega_1$ are kept distinct (~2 % difference in LiNbO$_3$).

3. Theory

Two-mode TCMT

Pump at ω_1, slowly varying amplitudes $a_n(t)$ such that $|a_n|^2$ = stored modal energy:

$$\boxed{\frac{da_2}{dt} = \left(i\Delta\omega - \tfrac{\gamma_2}{2}\right)a_2 + i\beta\,a_1^2}$$

$$\boxed{\frac{da_1}{dt} = -\tfrac{\gamma_1}{2}a_1 + i\beta^*\,a_1^*\,a_2 + i\alpha_3\,|a_1|^2a_1 + \sqrt{\gamma_1^{(\text{rad})}}\,s_+}$$

with **input-output** $s_- = -s_+ + \sqrt{\gamma_1^{(\text{rad})}}\,a_1$.

The **single coupling constant** β appears in both SHG and DFG terms — this is Manley–Rowe / energy conservation. Numerically:

$$\beta = \frac{\varepsilon_0\,\omega_1}{4}\int \chi^{(2)}_{ijk}(\mathbf{r})\,f^*_{2,i}\,f_{1,j}\,f_{1,k}\,d^3r$$

The bare Kerr:

$$\alpha_3 = \frac{3\varepsilon_0\omega_1}{8}\int \chi^{(3)}_{ijkl}f^*_{1,i}f_{1,j}f_{1,k}f^*_{1,l}d^3r$$

Adiabatic elimination → effective $\chi^{(3)}$

Assume $|\Delta\omega| \gtrsim 2\gamma_2$ and $|\dot{a}_2| \ll |\Delta\omega\,a_2|$. Setting $da_2/dt \to 0$:

$$a_2 = \frac{-\beta\,a_1^2}{\Delta\omega + i\gamma_2/2}$$

Substituting into the FM equation,

$$\frac{da_1}{dt} = -\tfrac{\gamma_1}{2}a_1 - \underbrace{\frac{|\beta|^2\gamma_2/2}{\Delta\omega^2 + (\gamma_2/2)^2}}_{\text{cascade TPA (real)}}|a_1|^2a_1 + i\underbrace{\left[\alpha_3 - \frac{|\beta|^2\,\Delta\omega}{\Delta\omega^2 + (\gamma_2/2)^2}\right]}_{\text{effective Kerr}}|a_1|^2a_1 + \sqrt{\gamma_1^{(\text{rad})}}\,s_+$$

The cascade contributes both a **Kerr-like** real coefficient and a **TPA-like** absorptive coefficient, both Lorentzian in $\Delta\omega$.

The two cascaded coefficients

Cascaded Kerr (dispersive, sign-tunable):

$$\alpha_3^{\text{casc}}(\Delta\omega) = -\frac{|\beta|^2\,\Delta\omega}{\Delta\omega^2 + (\gamma_2/2)^2}$$

Limits:

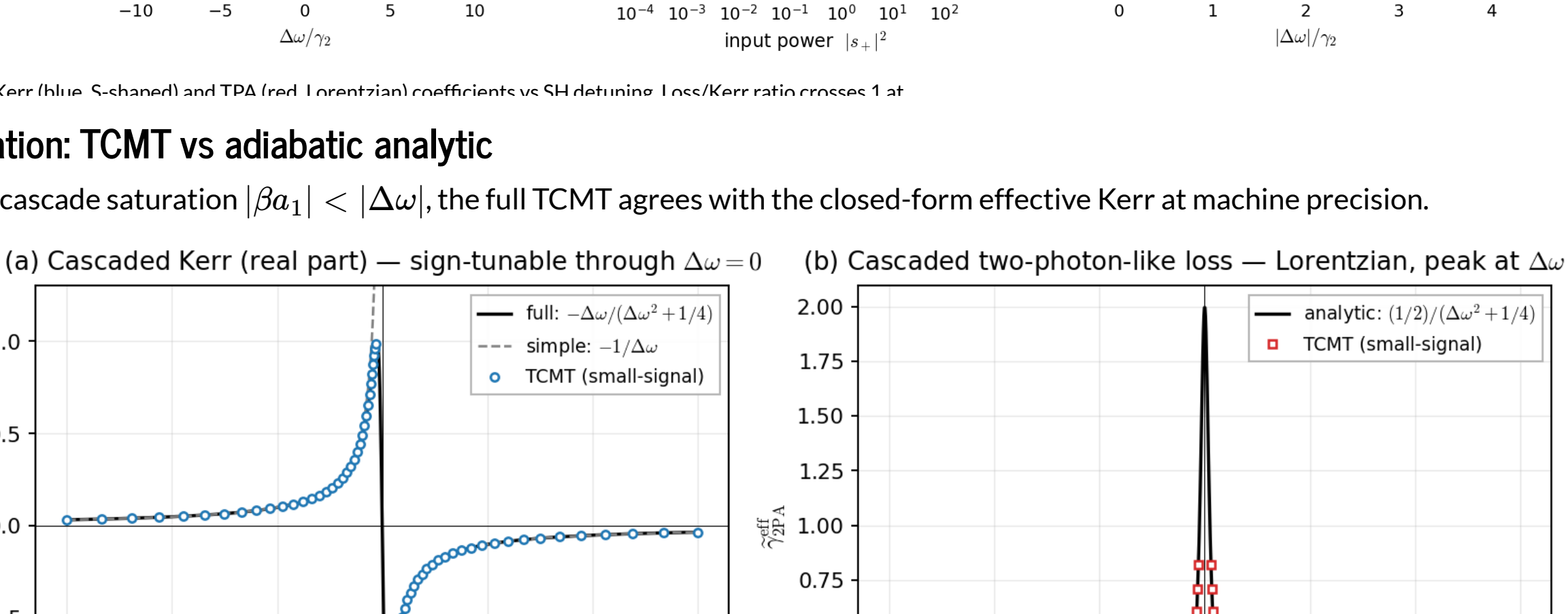
- $|\Delta\omega| \gg \gamma_2/2$: $\quad \alpha_3^{\text{casc}} \to -|\beta|^2/\Delta\omega$
- Sign flips at $\Delta\omega = 0$ — self-focusing ↔ self-defocusing on demand.

Cascaded TPA (absorptive):

$$\gamma_3^{\text{casc}}(\Delta\omega) = \frac{|\beta|^2\,\gamma_2/2}{\Delta\omega^2 + (\gamma_2/2)^2}$$

Limits:

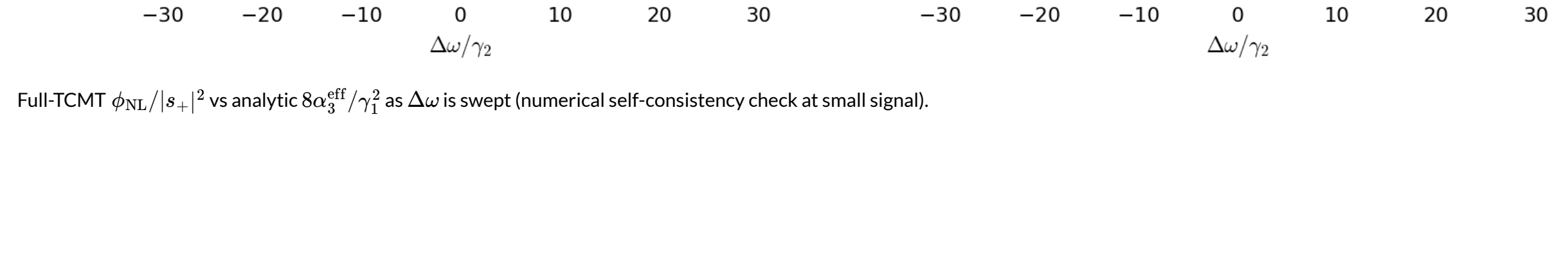
- Peak at $\Delta\omega = 0$ with magnitude $2|\beta|^2/\gamma_2 \propto Q_2$.
- $|\Delta\omega| \gg \gamma_2$: TPA suppressed as $1/\Delta\omega^2$.



Cascade Kerr (blue, S-shaped) and TPA (red, Lorentzian) coefficients vs SH detuning. Loss/Kerr ratio crosses 1 at

Validation: TCMT vs adiabatic analytic

Below cascade saturation $|\beta a_1| < |\Delta\omega|$, the full TCMT agrees with the closed-form effective Kerr at machine precision.



Full-TCMT $\phi_{\text{NL}}/|s_+|^2$ vs analytic $8\alpha_3^{\text{eff}}/\gamma_1^2$ as $\Delta\omega$ is swept (numerical self-consistency check at small signal).

Bandwidth-enhancement sum rule

Peak enhancement and useful bandwidth trade against each other:

$$\boxed{\chi^{(3)}_{\text{casc}\;\text{peak}} \cdot \Delta f_{\text{usable}} \;\lesssim\; \frac{\omega_1\,|\chi^{(2)}|^2\,\eta_{\text{geom}}}{12\pi\,n_2^2} \quad (\text{Q}_2\text{-independent})}$$

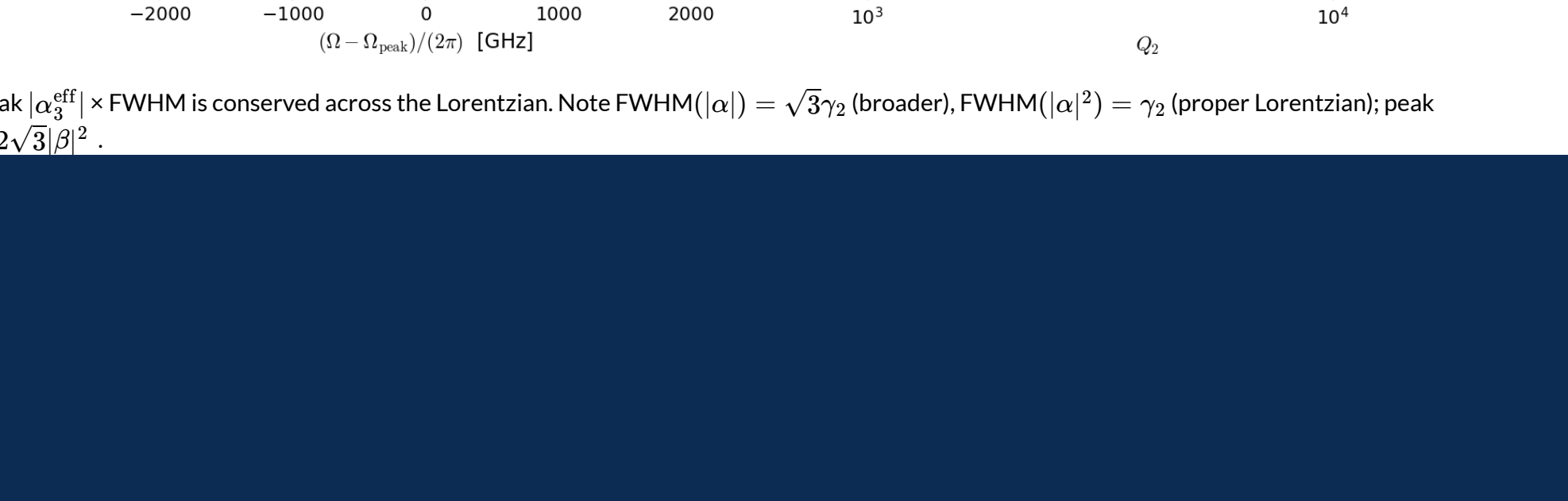
Reading: Q_2 pulls a tall narrow Lorentzian out of a fixed area; you cannot beat the sum rule by pushing Q_2.

This is the **same rule** that constrains resonant linear absorption (oscillator-strength sum rule) — purely classical.

At $Q_2 = 4{,}000$ in LiNbO$_3$:

- full SH linewidth $\gamma_2/(2\pi) \approx 97\,\mathrm{GHz}$
- usable bandwidth $\sim$ same

Narrowband by Kerr-electronics standards, broadband by photonic-cavity standards.



Sum-rule constancy: peak $|\alpha_3^{\text{eff}}|$ × FWHM is conserved across the Lorentzian. Note $\text{FWHM}(|\alpha|) = \sqrt{3}\gamma_2$ (broader), $\text{FWHM}(|\alpha|^2) = \gamma_2$ (proper Lorentzian); peak $|\alpha| \times \text{FWHM}(|\alpha|) = 2\sqrt{3}|\beta|^2$.

4. Comparison with native $\chi^{(3)}$ resonance

The cascade enhancement ratio R

Define

$$\boxed{R \equiv \frac{|\alpha_3^{\text{casc}}|}{|\alpha_3^{\text{native}}|} = \frac{|\beta|^2}{|\alpha_3|\,|\Delta\omega + i\gamma_2/2|} \overset{|\Delta\omega|\gg\gamma_2}{\simeq} \frac{|\chi^{(3)}_{\text{casc}}|}{|\chi^{(3)}_{\text{native}}|}}$$

For LiNbO$_3$ at the worked operating point ($Q_2 = 4{,}000$, $\Delta\omega = 2\gamma_2$, $\eta_{\text{geom}} = 0.3$):

$$|\chi^{(3)}_{\text{casc}}| \sim \frac{Q_2}{24\,n_2^2}\,|\chi^{(2)}|^2\,\eta_{\text{geom}} \simeq 2.5 \times 10^{-20}\,\mathrm{m}^2/\mathrm{V}^2 \Rightarrow \mathbf{R \approx 12}$$

ⓘ **Note**

Pushing $\Delta\omega \to \gamma_2$ (edge of adiabatic validity) raises R to ~20 (full Lorentzian — the often-quoted "$R \sim 25$" is from the simple $1/\Delta\omega$ approximation, which overshoots by ~20% at $\Delta\omega = \gamma_2$). Loss/Kerr ratio there is $\gamma_2/(2|\Delta\omega|) = 1/2$ — TPA is approximately equal to the cascade-Kerr, a substantial operational penalty.

SPM phase shift — leverage $(1 + R)$

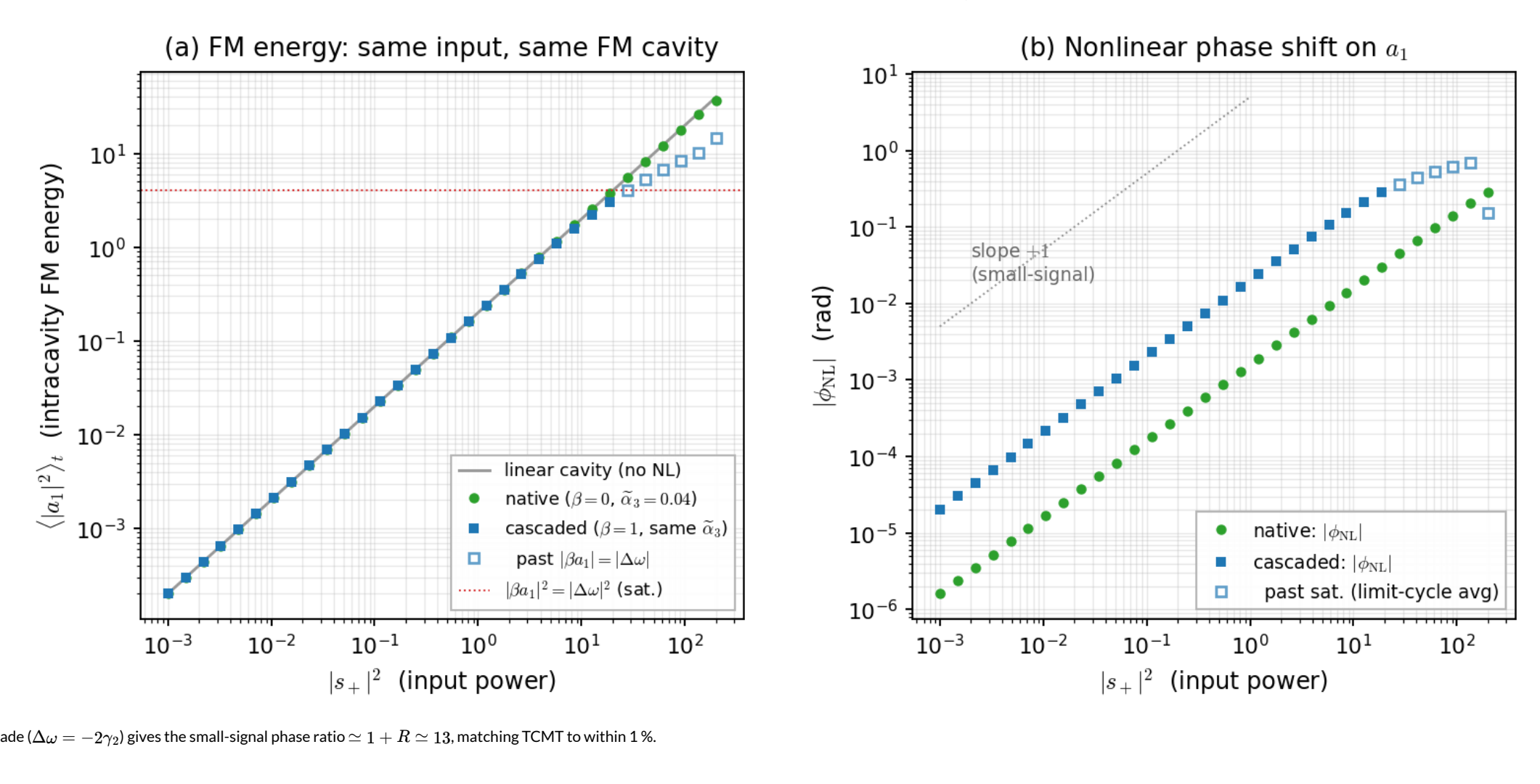
Steady-state nonlinear phase per input photon at on-resonance critical-coupled pump:

$$\frac{\phi_{\rm NL}}{|s_+|^2} = \frac{8\,\alpha_3^{\rm eff}}{\gamma_1^2}$$

$$\boxed{\frac{\phi_{\rm NL}^{\rm A+B}}{\phi_{\rm NL}^{\rm A}} = 1 + R}$$

Validity: $|\beta a_1| \ll |\Delta\omega|$ (small-signal cascade); $|s_+|^2 \ll |\Delta\omega|^2(\gamma_1/2)^2/(\gamma_1^{\rm rad}\beta^2)$.

Native vs cascade-added -- same FM cavity, same input ($\tilde{\alpha}_3 = +0.04$, $\Delta\omega = -2\gamma_2$, constructive cascade; $R \simeq 12$, ratio $\simeq 1 + R \simeq 13$)


(a) FM energy: same input, same FM cavity

(b) Nonlinear phase shift on a_1

Constructive cascade ($\Delta\omega = -2\gamma_2$) gives the small-signal phase ratio $\simeq 1 + R \simeq 13$, matching TCMT to within 1 %.

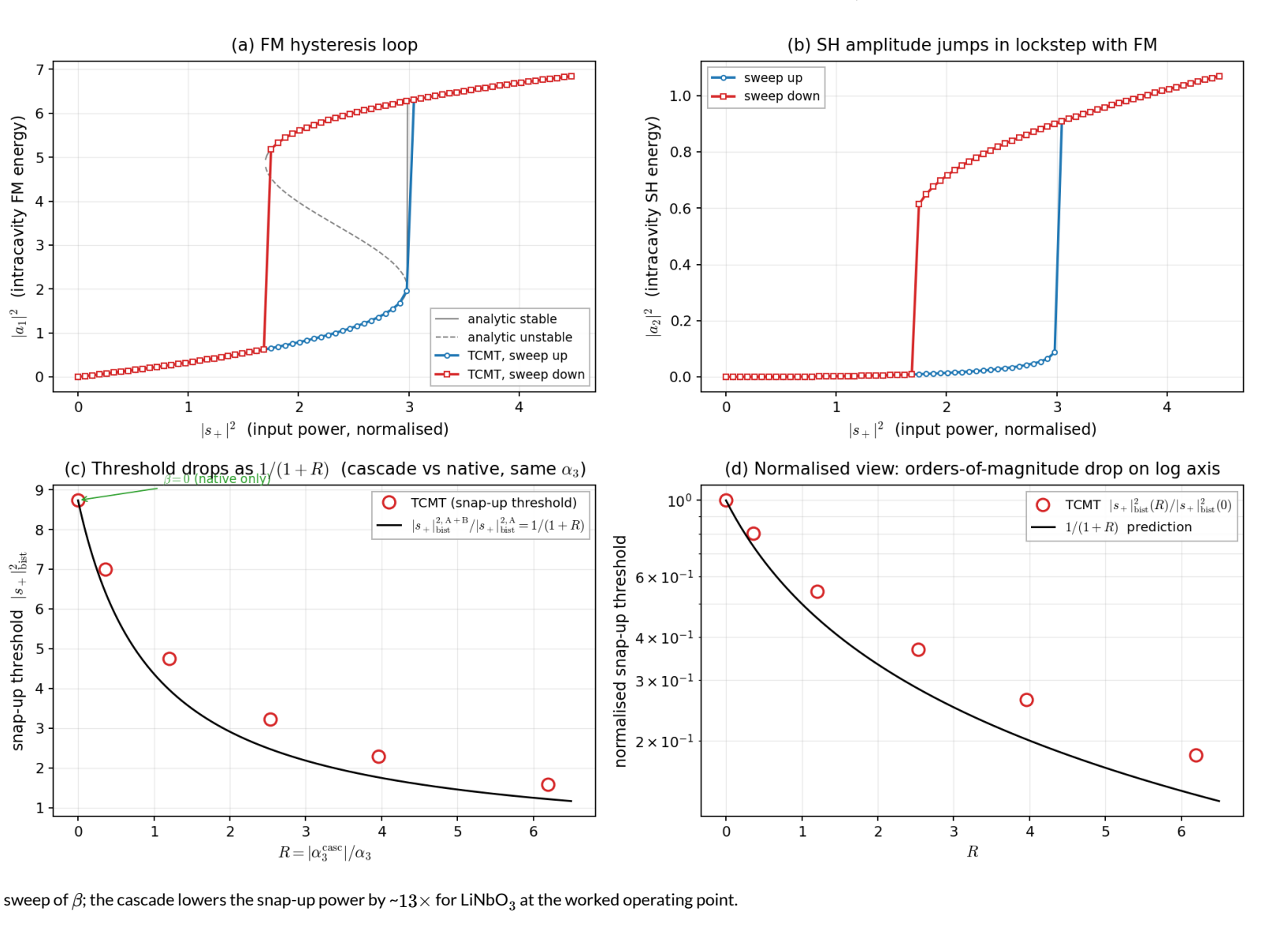
Bistability threshold — leverage $1/(1 + R)$

For a pump detuned past the bistability threshold $|\delta_p| > \sqrt{3}\,\gamma_1/2$ with $\mathrm{sign}(\delta_p) = -\mathrm{sign}(\alpha_3^{\rm eff})$:

$$|s_+|^2_{\rm bist} \;\propto\; \frac{\gamma_1^2}{|\alpha_3^{\rm eff}|} \;\Rightarrow\; \boxed{\frac{|s_+|^{2,\,\rm A+B}_{\rm bist}}{|s_+|^{2,\,\rm A}_{\rm bist}} = \frac{1}{1+R}}$$

Same factor controls the **all-optical switching energy**: $E_{\rm switch}^{\rm A+B}/E_{\rm switch}^{\rm A} = 1/(1 + R)$.

Cascaded Kerr bistability ($\Delta\omega/\gamma_2 = 5.0$, $\delta_1/\gamma_2 = 0.8$, blue-detuned). Panels (c,d) validate the $1/(1+R)$ threshold scaling across a range of R.


(a) FM hysteresis loop

(b) SH amplitude jumps in lockstep with FM

(c) Threshold drops as $1/(1+R)$ (cascade vs native, same α_3)

(d) Normalised view: orders-of-magnitude drop on log axis

Threshold reduction $1/(1 + R)$ validated across a sweep of β; the cascade lowers the snap-up power by ~$13\times$ for LiNbO$_3$ at the worked operating point.

Four-wave mixing — leverage $(1 + R)^2$

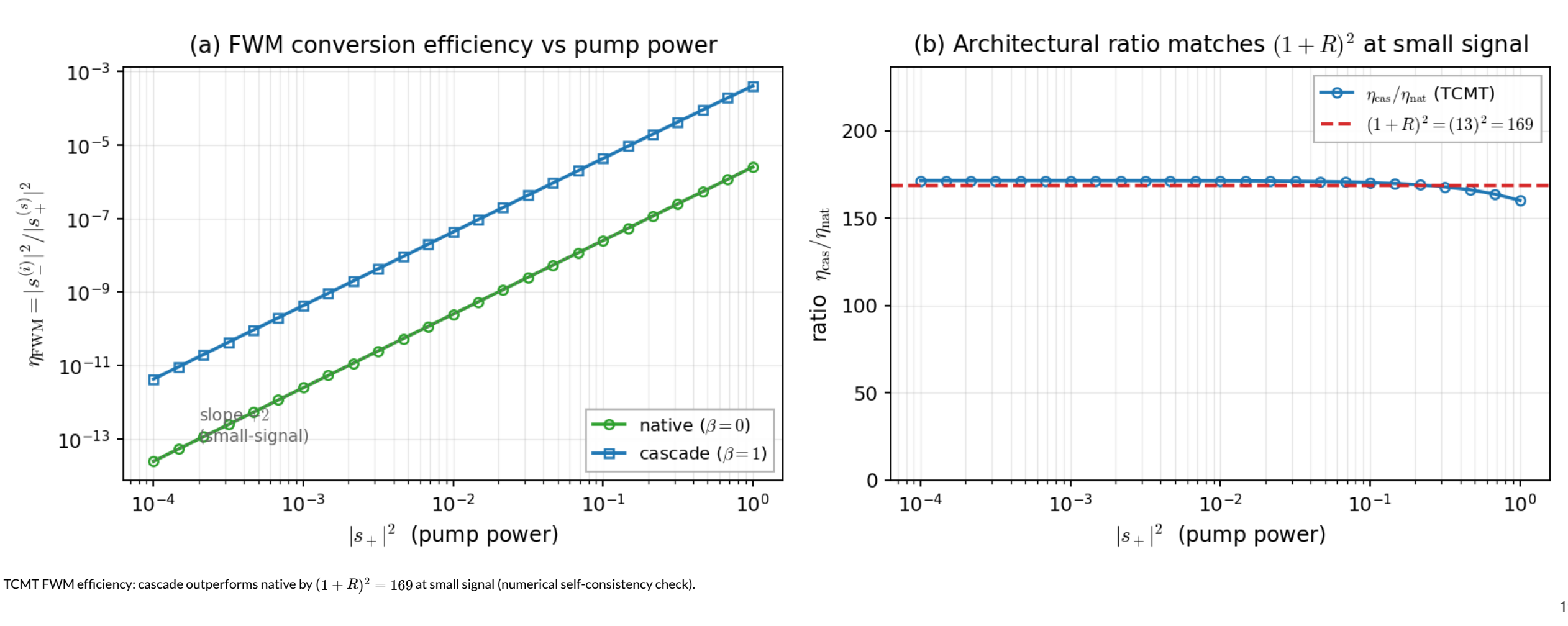
Pump at $\omega_p \approx \omega_1$ + weak signal at $\omega_s = \omega_p - \Omega$ → idler at $\omega_i = \omega_p + \Omega$ inside the FM linewidth.

$$\eta_{\rm FWM} \equiv \frac{|s_-^{(i)}|^2}{|s_+^{(s)}|^2} \propto \frac{|\alpha_3^{\rm eff}|^2 |a_1^{\rm pump}|^4}{(\gamma_1/2)^2}$$

$$\boxed{\frac{\eta_{\rm FWM}^{\rm A+B}}{\eta_{\rm FWM}^{\rm A}} = (1 + R)^2}$$

Validity: $|\Omega| < \min(\gamma_1, \gamma_2)$; small-signal cascade. $(1 + R)^2 = 169$ with $R = 12$ at the design point — the **most strongly differentiating** single-observable test.

FWM efficiency: cascade outperforms native by $(1+R)^2$ -- $\Omega/\gamma_2 = 0.5$, $\tilde{\alpha}_3 = +0.04$, $\Delta\omega = -2\gamma_2$ (constructive; $R \simeq 12$)


(a) FWM conversion efficiency vs pump power

(b) Architectural ratio matches $(1+R)^2$ at small signal

TCMT FWM efficiency: cascade outperforms native by $(1 + R)^2 = 169$ at small signal (numerical self-consistency check).

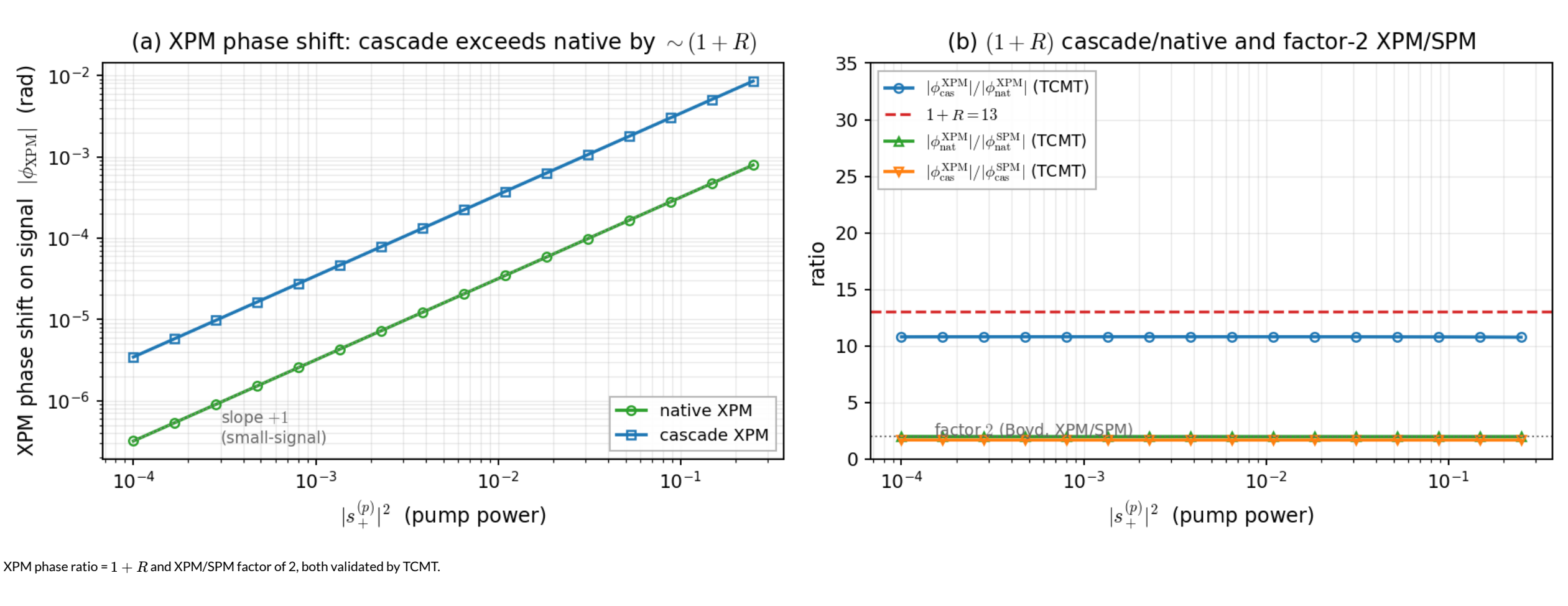
Cross-phase modulation

Separate weak signal beam (distinguishable polarization or frequency offset) → refractive-index shift driven by pump intensity $|a_1^{\rm pump}|^2$:

$$\boxed{\frac{\alpha_3^{\rm XPM,\,A+B}}{\alpha_3^{\rm XPM,\,A}} = 1 + R, \quad \alpha_3^{\rm XPM} = 2\,\alpha_3^{\rm SPM}}$$

Tensor-channel caveat: $1 + R$ holds for tensor contractions where both cascade and native channels are non-vanishing. In mixed-polarization channels (e.g., $e \to \mathrm{SH} \to o$) the cascade XPM uses different $\chi_{ijk}^{(2)}$ entries than the native XPM channel — verify for the experimental geometry.

XPM: cascade $\times(1+R)$ over native; XPM $\times 2$ over SPM -- $\Omega = 0.5\gamma_2$, $\tilde{\alpha}_3 = +0.04$, $\Delta\omega = -2\gamma_2$ (constructive; $R \simeq 12$)


(a) XPM phase shift: cascade exceeds native by $\sim(1+R)$

(b) $(1+R)$ cascade/native and factor-2 XPM/SPM

XPM phase ratio = $1 + R$ and XPM/SPM factor of 2, both validated by TCMT.

Sign tunability — *unique to cascade*

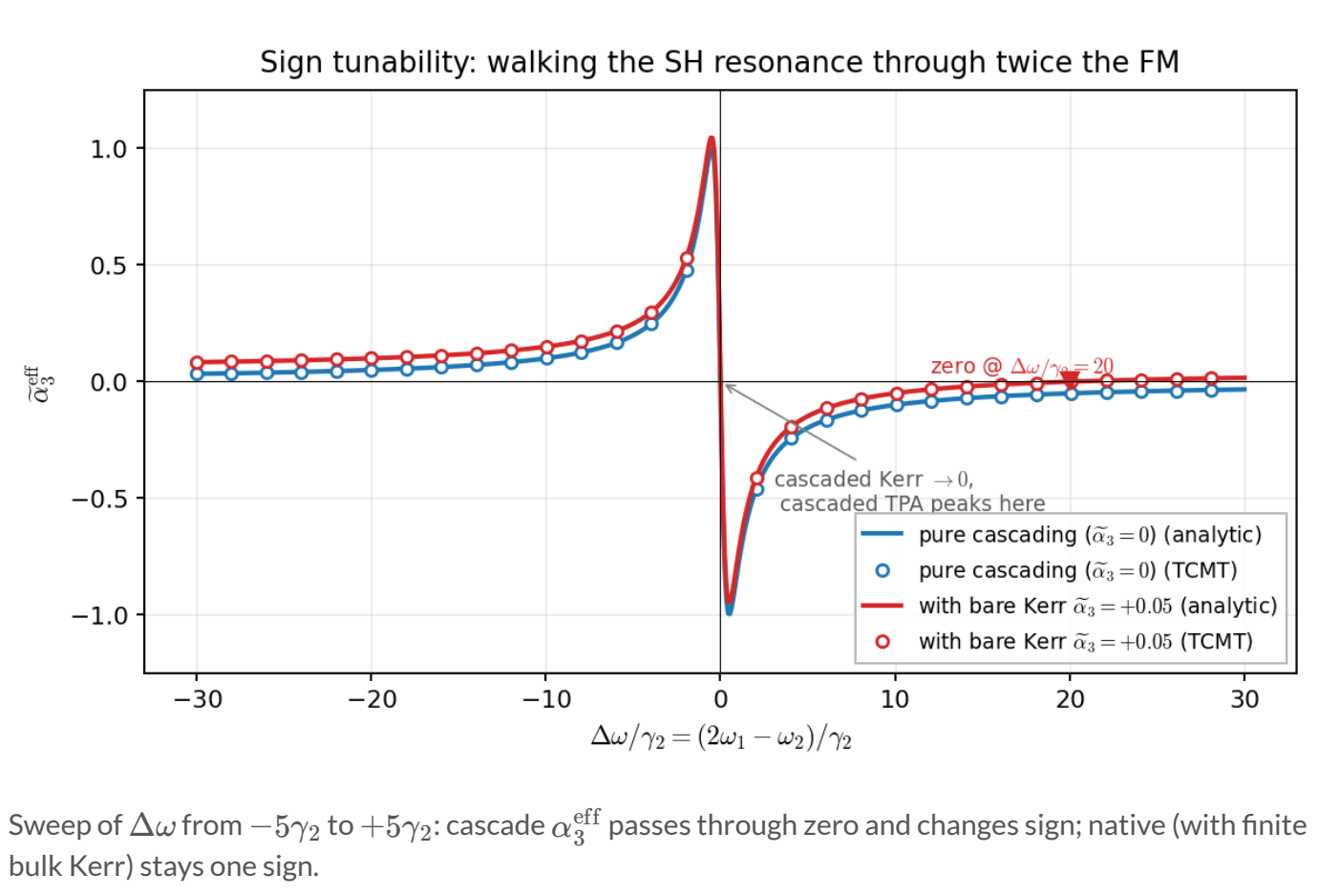
$\alpha_3^{\rm casc}(\Delta\omega) = -|\beta|^2\Delta\omega/(\Delta\omega^2 + (\gamma_2/2)^2)$ **changes sign** at $\Delta\omega = 0$:

- $\Delta\omega > 0$ → self-defocusing
- $\Delta\omega < 0$ → self-focusing
- $\Delta\omega = 0$ → pure-loss (TPA only)

A single device can be tuned thermo-optically, electro-optically (Pockels on the SH cavity), or by strain — no fabrication swap required.

Native $\chi^{(3)}$ has **no analogue** at any Q_1.

Sign tunability: walking the SH resonance through twice the FM



Sweep of $\Delta\omega$ from $-5\gamma_2$ to $+5\gamma_2$: cascade $\alpha_3^{\rm eff}$ passes through zero and changes sign; native (with finite bulk Kerr) stays one sign.

5. Cascade-only capabilities

Saturable absorber (cascaded TPA)

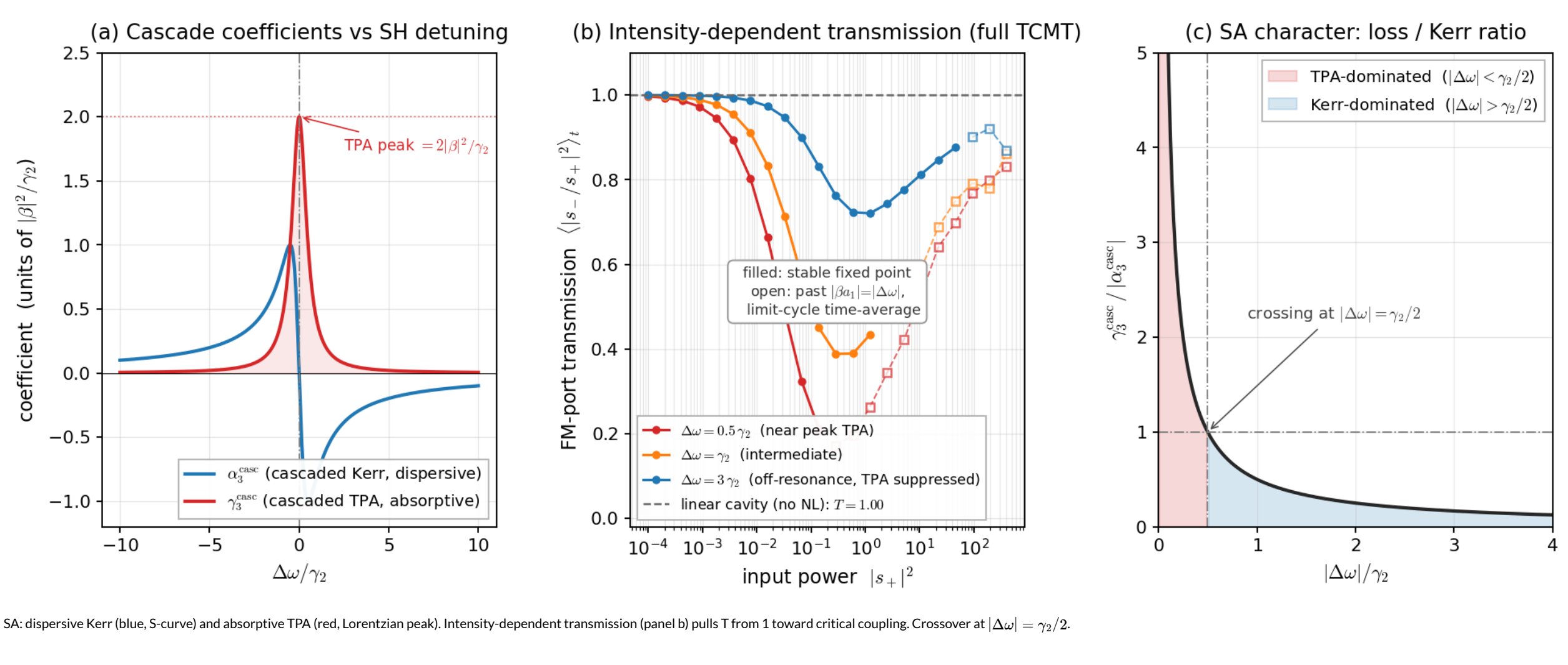
Operating near $\Delta\omega = 0$, the cascaded coefficient is **pure imaginary** — Lorentzian-peaked intensity-dependent loss:

$$\gamma_3^{\rm casc}(\Delta\omega = 0) = \frac{2|\beta|^2}{\gamma_2} \;\propto\; Q_2$$

Detunable unlike intrinsic semiconductor TPA: shift ω_2 by $\gtrsim$ a few γ_2 to turn the SA off ($\propto 1/\Delta\omega^2$).

Recovery time $\sim 1/\gamma_2 \sim 1\,$ps at $Q_2 = 4{,}000$ — faster than SESAM excited-state lifetimes, all-optical, no thermal drift.

Cascaded saturable absorber: $\Delta\omega$ tunes between absorptive (TPA) and dispersive (Kerr) cascade response


(a) Cascade coefficients vs SH detuning

(b) Intensity-dependent transmission (full TCMT)

(c) SA character: loss / Kerr ratio

SA: dispersive Kerr (blue, S-curve) and absorptive TPA (red, Lorentzian peak). Intensity-dependent transmission (panel b) pulls T from 1 toward critical coupling. Crossover at $|\Delta\omega| = \gamma_2/2$.

AM modulator via $\Delta\omega(t)$

Modulate the SH-cavity detuning (electro-optic / thermo-optic / piezo): $\Delta\omega(t) = \Delta\omega_0 + \delta\Omega\sin(\Omega_{\rm mod}t)$ → modulates $\alpha_3^{\rm casc}$ → modulates FM-port transmission at fixed pump.

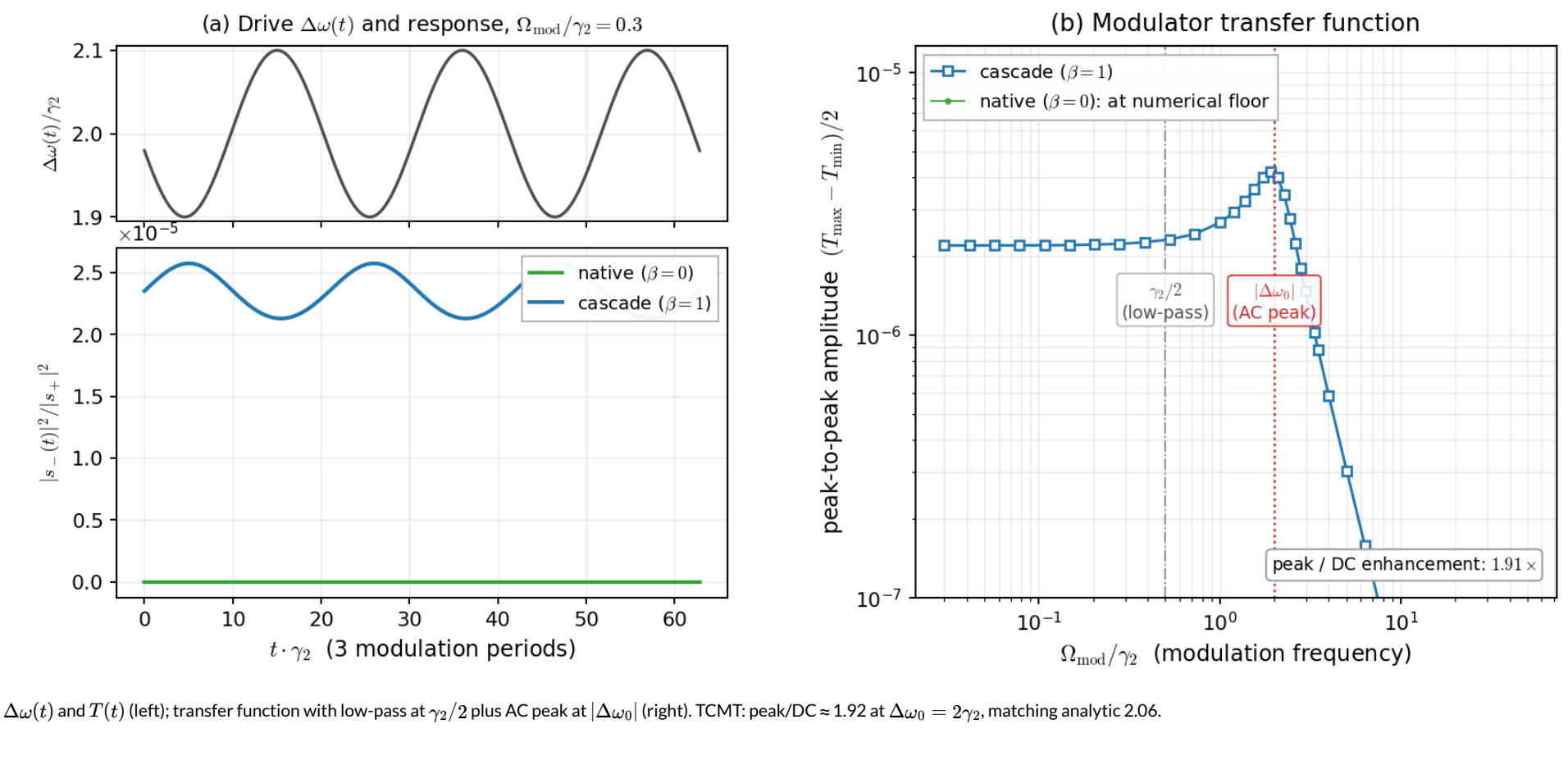
Two transfer-function features (from linearised time-dependent SH eq.):

- **Low-pass roll-off** at $\Omega_{\rm mod} \sim \gamma_2/2$ (−3 dB at $\gamma_2/(4\pi) \approx 50$ GHz, Q_2=4,000). Full SH linewidth $\gamma_2/(2\pi) \approx 100$ GHz is the *intrinsic* cascade BW, not the modulator −3 dB.
- **AC resonant peak** at $\Omega_{\rm mod} \approx |\Delta\omega_0|$ (validity: $|\Delta\omega_0| \gg \gamma_2/2$). Peak/DC ratio of the $\alpha_3^{\rm eff}$ modulation amplitude is

$$\frac{|\delta\alpha_3^{\rm eff}|_{\rm peak}}{|\delta\alpha_3^{\rm eff}|_{\rm DC}} = \frac{\sqrt{\Delta\omega_0^2 + (\gamma_2/2)^2}}{\gamma_2} \xrightarrow[\Delta\omega_0 \gg \gamma_2]{} \frac{|\Delta\omega_0|}{\gamma_2} \simeq 2 \text{ at } \Delta\omega_0 = 2\gamma_2.$$

No native-Kerr analogue: bulk $\chi^{(3)}$ has no detuning knob. (Electro-optic Pockels modulators exist as a separate class and routinely hit > 100 GHz — they modulate the *linear* index, not α_3. The cascade is uniquely a Kerr-coefficient modulator via a secondary cavity.)

AM modulator via $\Delta\omega(t)$ ($\Delta\omega_0 = 2\gamma_2$, $\delta\Omega = 0.1\,\gamma_2$, $s_+ = 0.5$)


(a) Drive $\Delta\omega(t)$ and response, $\Omega_{\rm mod}/\gamma_2 = 0.3$

(b) Modulator transfer function

Time-domain co-modulation of $\Delta\omega(t)$ and $T(t)$ (left); transfer function with low-pass at $\gamma_2/2$ plus AC peak at $|\Delta\omega_0|$ (right). TCMT: peak/DC ≈ 1.92 at $\Delta\omega_0 = 2\gamma_2$, matching analytic 2.06.

Passive optical limiter

Cascade saturation $|\beta a_1| \to |\Delta\omega|$ is a **parametric intensity ceiling**:

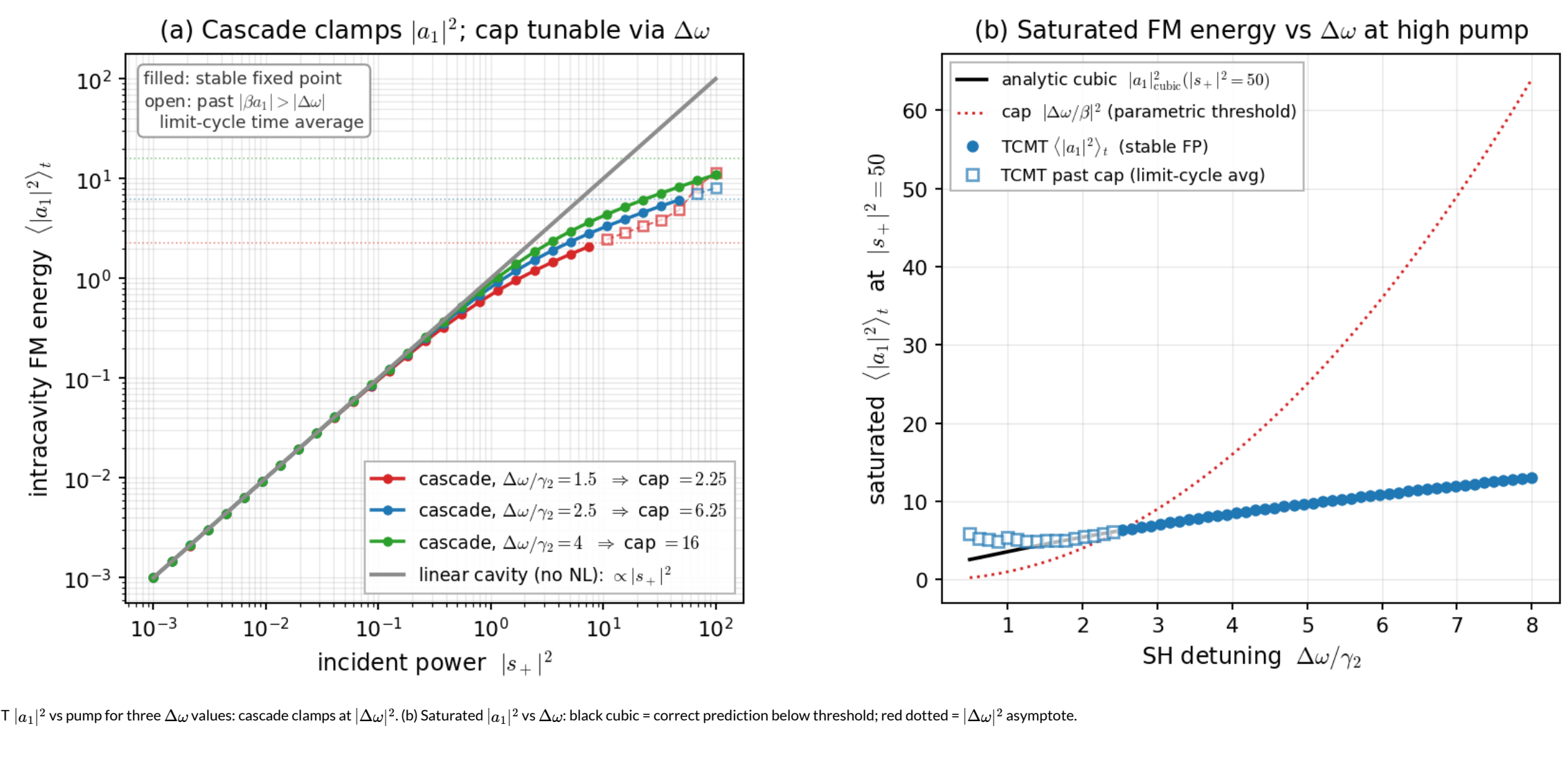
$$\boxed{|a_1|^2_{\rm cap} = |\Delta\omega/\beta|^2 \;\propto\; (1/Q_2)^2}$$

Above this threshold the trivial fixed point destabilises (OPO threshold) and the FM clamps — excess pump dissipates through cascaded TPA into the heat-sunk SH mode.

Tunable ceiling: sweep $\Delta\omega$ quadratically to set the clamp level (laser-safety analogue, all-optical, no thermal load on the FM port).

What's different from native materials: native dielectrics *do* have passive intensity limits (TPA in semiconductors, free-carrier absorption near band edge, thermal effects in any high-Q cavity). The distinguishing feature of the cascade is that the clamp is **parametric** (oscillation-threshold) rather than dissipative — heat-sunk into the SH mode rather than thermalised in the FM-port material.

Cascade as a passive optical limiter: intensity-clamped at $|a_1|^2_{\rm cap} = |\Delta\omega/\beta|^2$


(a) Cascade clamps $|a_1|^2$; cap tunable via $\Delta\omega$

(b) Saturated FM energy vs $\Delta\omega$ at high pump

(a) Full-TCMT $|a_1|^2$ vs pump for three $\Delta\omega$ values: cascade clamps at $|\Delta\omega|^2$. (b) Saturated $|a_1|^2$ vs $\Delta\omega$: black cubic = correct prediction below threshold; red dotted = $|\Delta\omega|^2$ asymptote.

$\Delta\omega$-readout transducer — sensor

Any perturbation shifting ω_2 (temperature, strain, adsorbed molecules, Pockels via DC field) → shifts $\Delta\omega$ → shifts $\alpha_3^{\rm casc}$ → shifts $\phi_{\rm NL}$.

Lorentzian-derivative response, Q_2^2 scaling

$$\frac{d\alpha_3^{\rm casc}}{d\Delta\omega} = -|\beta|^2\frac{(\gamma_2/2)^2 - \Delta\omega^2}{(\Delta\omega^2 + (\gamma_2/2)^2)^2}$$

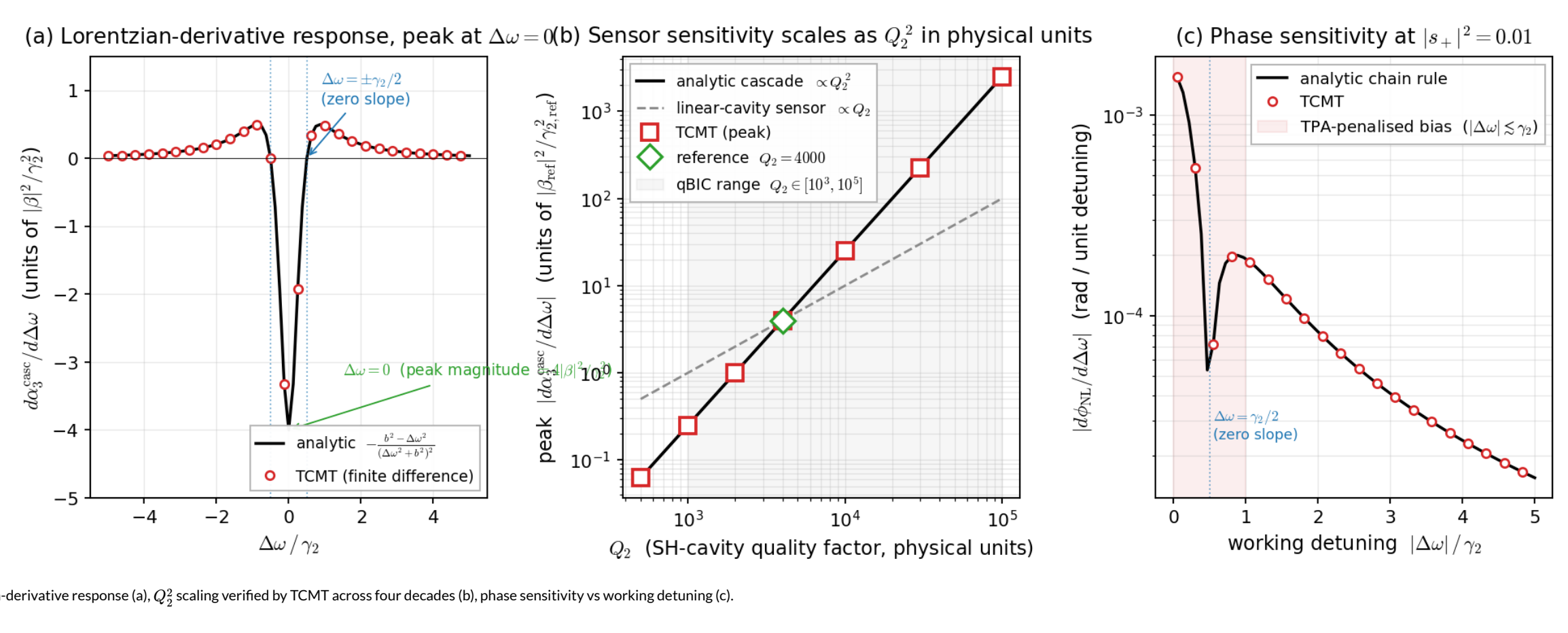
Peak at $\Delta\omega = 0$:

$$\boxed{\left.\frac{d\alpha_3^{\rm casc}}{d\Delta\omega}\right|_{\max} = \frac{4|\beta|^2}{\gamma_2^2} \;\propto\; \mathbf{Q_2^2}}$$

Two factors of Q_2: one from $|\alpha_3^{\rm casc}| \propto Q_2$, one from the Lorentzian-derivative slope $\propto 1/\gamma_2 \propto Q_2$.

Operating-point caveat (on the *same* slide): - Peak sensitivity is at $\Delta\omega = 0$, which is also the **peak of cascaded TPA loss**. The most sensitive bias is also the lossiest. - Honest figure of merit: $\mathrm{slope}/\sqrt{\gamma_3^{\rm eff}}$. The cascade's Q_2^2 scaling becomes $Q_2^{3/2}$ once shot-noise from TPA-dissipated power is folded in — still better than linear-cavity ($Q_2^{1/2}$ in this FoM), but **not the naïve** Q_2^2. - Practical bias: a few γ_2 off-resonance → $O(1)$ slope drop, $1/\Delta\omega^2$ TPA suppression. Q_2^2 scaling survives this trade-off; prefactor reduces.

$\Delta\omega$-readout transducer: peak at $\Delta\omega = 0$, Q_2^2 scaling beats linear-cavity Q_2 sensors


(a) Lorentzian-derivative response, peak at $\Delta\omega = 0$

(b) Sensor sensitivity scales as Q_2^2 in physical units

(c) Phase sensitivity at $|s_+|^2 = 0.01$

Lorentzian-derivative response (a), Q_2^2 scaling verified by TCMT across four decades (b), phase sensitivity vs working detuning (c).

6. Feasibility

Cascade vs native: who wins, by host

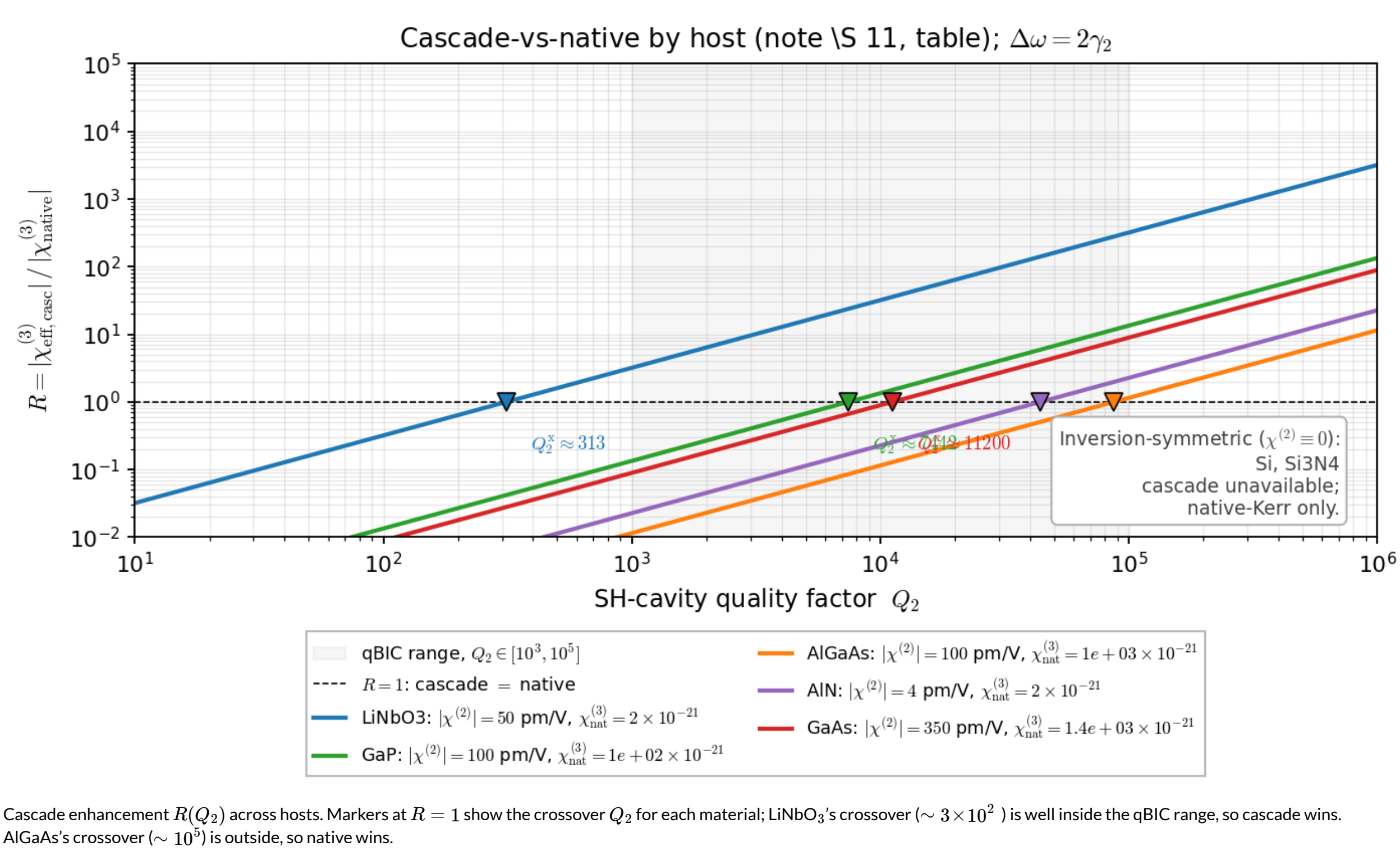
R at the conservative design point ($Q_2 = 4{,}000$, $\eta_{\rm geom} = 0.3$ assumed for *all materials*, $\Delta\omega = 2\gamma_2$):

Host	$\lvert\chi^{(2)}\rvert$ [m/V]	$\lvert\chi^{(3)}_{\rm native}\rvert$ [m²/V²]	R at Q_2=4×10^3	$Q_2^{\rm crossover}$
LiNbO$_3$	5×10^{-11}	2×10^{-21}	~ 12	$\sim 3\times10^2$
GaP	1×10^{-10}	10^{-19}	~ 0.5	$\sim 8\times10^3$
AlGaAs (near band-edge)	10^{-10}	10^{-18}	~ 0.05	$\sim 9\times10^4$
AlN	4×10^{-12}	2×10^{-21}	~ 0.1	$\sim 4\times10^4$
Si, Si$_3$N$_4$	$\equiv 0$	–	–	– (no cascade)

Caveat: all R values carry $\pm$ factor-of-2 uncertainty: $\eta_{\rm geom}$ is a fabrication-dependent overlap that may not reach 0.3 in every material (especially semiconductors with band-edge proximity); $|\chi^{(3)}_{\rm native}|$ has 50_3 wins; AlGaAs native wins) is robust.

Cascade-vs-native by host material

Cascade-vs-native by host (note \S 11, table); $\Delta\omega = 2\gamma_2$



Cascade enhancement $R(Q_2)$ across hosts. Markers at $R = 1$ show the crossover Q_2 for each material; LiNbO$_3$'s crossover ($\sim 3\times10^2$) is well inside the qBIC range, so cascade wins. AlGaAs's crossover ($\sim 10^5$) is outside, so native wins.

Caveats specific to cascade

1. **Cascade-specific loss.** TPA-like coefficient $\gamma_3^{\rm eff} = |\beta|^2\gamma_2/(2\Delta\omega^2)$ has no analogue in low-loss native LiNbO$_3$. At fixed phase shift, cascade carries a loss-to-Kerr ratio of $\gamma_2/(2|\Delta\omega|)$.
2. **Lower saturation.** Cascade saturates at $|\beta a_1| \sim |\Delta\omega|$; max stored energy $\propto 1/Q_2^2$. Native saturates at thermal / damage limits, orders of magnitude higher.
3. **Fabrication complexity.** Native needs *one* high-Q resonance; cascade needs *two*, co-localised, with high spatial overlap.
4. **Speed.** Native Kerr is $\sim$ fs; cascade has bandwidth γ_2 ($\sim$ ps). Cascade is locked out of ultrafast applications by its own sum rule.

Disorder / inhomogeneous broadening

For qBIC SH cavities, $Q_2 \propto 1/\alpha^2$ where α is the meta-atom asymmetry parameter. Geometric disorder across the array broadens the **array-averaged** SH resonance below the per-element Q_2:

$$Q_2^{\rm array} \;\sim\; \frac{Q_2^{\rm element}}{1 + (Q_2^{\rm element}\cdot\sigma_\alpha/\alpha)^2}$$

The single largest practical risk of the doubly-resonant scheme.

For the headline $R \approx 12$ at $Q_2 = 4{,}000$, you need $Q_2^{\rm array} \geq 4{,}000$, i.e., **asymmetry** $\sigma_\alpha/\alpha \lesssim 1\%$ across the pump spot.

This is much harder than achieving $Q_2 = 4{,}000$ on a *single* meta-atom — it requires uniformity in sub-nm fabrication of the symmetry-breaking feature (etch depth, hole position, sidewall angle) across the array.

Mitigations:

1. **Smaller array footprint** with single-shot e-beam → tight uniformity.
2. **Spatial selectivity**: probe only a sub-region with matched resonance (lattice-disorder spectroscopy).
3. **Adaptive tuning**: thermo-optic uniformity across the array (resistive heaters per meta-atom).
4. **Lower-Q operating point**: $Q_2 = 1{,}000$ is more disorder-tolerant; gives $R \sim 3$ for LiNbO$_3$ — still wins over native.

? estimate achievable σ_α/α for the fabrication process.

Dynamic range: only ~1 decade window of usable pump

Operating constraints:

- **Floor (SNR):** need at least $\phi_{\mathrm{NL}} \gtrsim 10\,\mathrm{mrad}$ to read out with photodiodes / heterodyne. With $\alpha_3^{\mathrm{eff}} \sim 0.05$ (LiNbO$_3$) and $\gamma_1 \sim 100$ GHz: floor at $|s_+|^2 \sim 10^{-2}$ in dim TCMT units.
- **Ceiling (saturation):** parametric instability at $|\beta a_1| = |\Delta\omega|$, i.e., $|a_1|^2_{\mathrm{sat}} = |\Delta\omega|^2 \propto 1/Q_2^2$. At $\Delta\omega = 2\gamma_2, Q_2 = 4{,}000$: $|s_+|^2_{\mathrm{sat}} \sim 10^{-1}$ in same units.

→ **only ~1 decade between floor and ceiling**.

Pushing Q_2 higher *simultaneously* squeezes the ceiling down quadratically. A $Q_2 = 10^5$ design has $R \approx 30$, but a 10^{-3}-decade pump window — operationally unusable.

Implication: there is a “sweet Q_2” where the cascade leverage is large enough to detect, but the dynamic range is still operational.

For LiNbO$_3$: - $Q_2 = 1{,}000$: $R \approx 3$, 3-decade pump window - $Q_2 = 4{,}000$: $R \approx 12$, 1-decade pump window - $Q_2 = 10{,}000$: $R \approx 30$, 0.3-decade pump window - $Q_2 = 10^5$: $R \approx 300$, 0.03-decade — not usable

The conservative $Q_2 = 4{,}000$ design is **already at the edge** of the practical dynamic-range trade-off.

For applications that require the full R leverage simultaneously with multi-decade dynamic range, a $\chi^{(2)}$-active host with **larger** $|\chi^{(2)}|$ (boosting $|\beta|$, hence R at fixed Q_2) is the right move, *not* higher Q_2.

7. Key design parameters

Hierarchy of design knobs

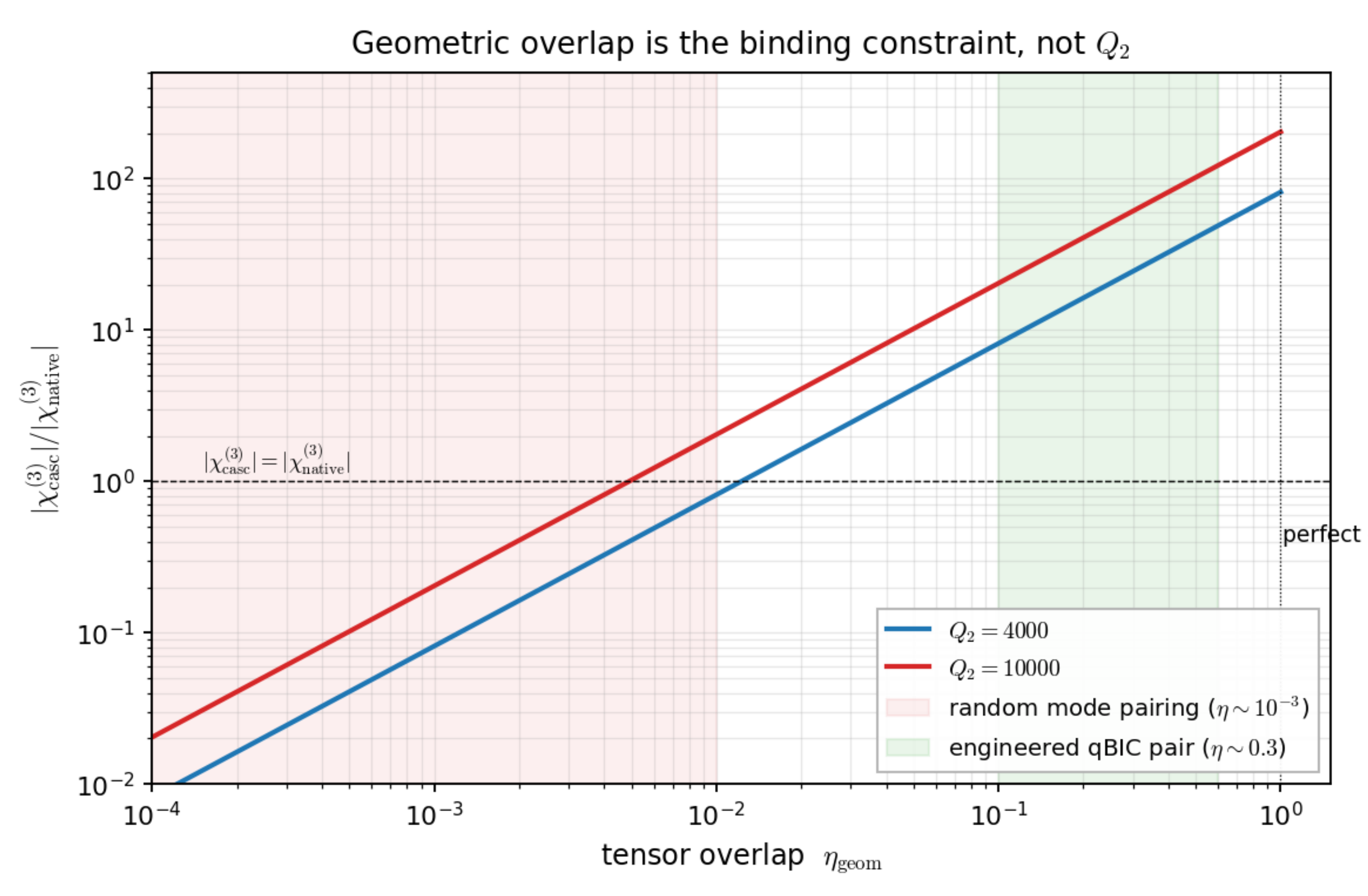
In order of decreasing leverage on $|\chi^{(3)}_{\mathrm{casc}}|$:

1. η_{geom} — the dimensionless modal overlap

$$\eta_{\mathrm{geom}} \equiv \frac{|\int \chi^{(2)}_{ijk} f^*_{2,i} f_{1,j} f_{1,k} d^3r|^2/V}{\int |\mathbf{f}_1|^4 d^3r}$$

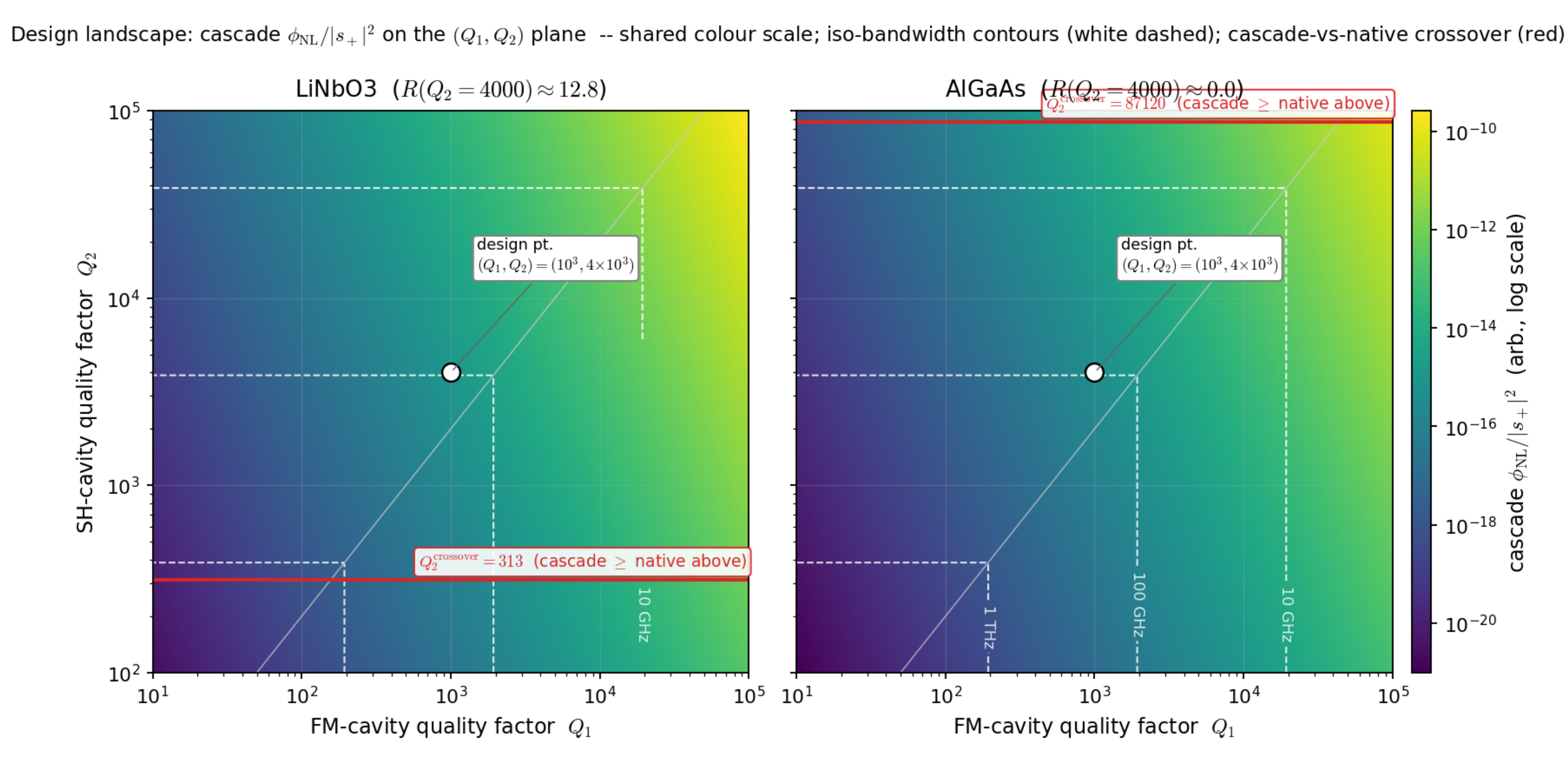
2. $|\chi^{(2)}|$ of the host — enters squared. LiNbO$_3$ vs AlGaAs is a factor 2.
3. $Q_2/n_2^2 - 1/n_2^2$ is fixed by material; Q_2 is the design knob.
   - Factor of 2 in Q_2 → factor of 2 in peak height.
   - Factor of 2 in Q_2 → factor of 2 cut in bandwidth (sum rule).
   - Factor of 2 in Q_2 → factor of 4 cut in saturation power.
4. $\Delta\omega$ **operating point** — controls Kerr-vs-TPA mix, sign, and validity. Three regimes:
   - $\Delta\omega = 2\gamma_2$: conservative, fully adiabatic, $R \approx 12$.
   - $\Delta\omega = \gamma_2$: edge of validity, $R \approx 20$ (full Lorentzian) but TPA/Kerr ratio $\sim 1/2$.
   - $\Delta\omega = 0$: pure SA, no Kerr.

Geometric overlap is the binding constraint



η_{geom} spans three orders of magnitude — random mode pairings give $\eta \sim 10^{-3}$; engineered qBIC pairs reach $\eta \sim 0.3$. The cascade ratio R scales linearly with η^{**}.

Design-space heatmap



Phase-bandwidth design map for LiNbO$_3$ as a function of (Q_1, Q_2). Diagonal contours = constant phase-bandwidth product (sum-rule lines). Practical qBIC range $Q_1, Q_2 \in [10^2, 10^5]$.

Practical trade-offs

You want…	Trade-off	Mitigation
Higher $\lvert\chi^{(3)}_{\mathrm{casc}}\rvert$	narrower BW; lower saturation	higher Q_2; active cooling
Wider BW	lower peak; more pump for same ϕ	moderate Q_2
Faster temporal response	loses cascade enhancement at $\sigma_t\gamma_2 \lesssim 1$	switch to bulk PPLN regime
Sign tunability w/o swap	fine $\Delta\omega$ control	thermo-optic / Pockels tuning
Avoid OPO threshold	operate below $\lvert\beta a_1\rvert \sim \lvert\Delta\omega\rvert$	larger $\Delta\omega$, lower pump
Avoid thermal confusion	absorption → heating	LiNbO$_3$ over GaAs
Suppress cascaded TPA	TPA peaks at $\Delta\omega = 0$	operate at $\lvert\Delta\omega\rvert \gtrsim 2\gamma_2$

8. Summary and asks

Summary of observable leverage

Observable	(A+B)/A ratio	Comment
SPM phase shift	$1 + R$	Foundational, $R \approx 12$ for LiNbO$_3$
XPM phase shift	$1 + R$	Same R; tensor-channel caveat
Bistability threshold	$1/(1 + R)$	Switching energy savings $\sim 13\times$
FWM conversion efficiency	$(1 + R)^2$	$\sim 170\times$ — most differentiating
Saturable absorber rate	n/a	Cascade-only, $\Delta\omega$-tunable
AM modulator response	n/a	Cascade-only, $$100 GHz
Optical-limiter clamp	$\sim (\Delta\omega/\beta)^2$	Cascade-only, passive ceiling
Sensor sensitivity	$\propto Q_2^2$	Cascade-only; beats linear-cavity Q_2

Which experiment first?

Not all applications are equal as “first demos”:

Application	Leverage	Verdict
FWM	$(1 + R)^2 \approx 170$	**Strongest cascade signature.** Clean cw measurement (heterodyne idler power). Narrowband ($\Omega < \gamma_2$) but that’s a feature not a bug for a demo.
Sensor (Q_2^2)	$\propto Q_2^2$	**Cleanest scaling claim.** Q_2^2 is unambiguous if measured across multiple Q_2’s. Caveat: lossy-bias operating point.
SPM	$1 + R$	Hardest to measure cleanly (phase noise, drift). Not differentiating from other Kerr cavities.
Bistability	$1/(1 + R)$	A $13\times$ threshold drop can come from many things (better Q_1, thermal). Not unambiguously a cascade signature.
SA	new capability	Competes with SESAMs at similar BW. Cascade tunability is the unique feature, hard to sell as a first demo.
AM modulator	new capability	Competes with > 100 GHz Pockels modulators. Cascade is qualitatively different physics but not better performance.
Optical limiter	new capability	Competes with TPA / free-carrier limiters. Cascade is parametric not dissipative — interesting but niche.

> Tip
>
> **Opinion:** the first experiment should be one whose positive outcome is simultaneously (1) unambiguous evidence of the cascade mechanism and (2) a real performance gain over the next-best alternative.
>
> **FWM and the sensor are the only two that satisfy both.** AM/SA/limiter are interesting capabilities, but a “first demo” of those competes with mature technologies that are likely to outperform on the metrics that matter.

Should we do this experimentally?

Pros

- ~order-of-magnitude $\chi^{(3)}$ enhancement on a *standard* low-loss host (LiNbO$_3$).
- Several capabilities (AM, SA, limiter, sensor) **native cannot match**.
- All in a chip-scale dielectric metasurface, no exotic materials.
- Builds on existing high-Q qBIC metasurface platforms — same fabrication.

Cons / open questions

- Doubly-resonant fabrication: Q_2 uniformity across the array matters.
- Cascade TPA penalty in low-loss hosts (no native analogue).
- Bandwidth $\sim$ 100 GHz — not for ultrafast applications.
- The “killer demo” depends on which capability we prioritise (FWM has highest leverage; sensor has cleanest theoretical claim).

Questions

1. **Target SH-mode platform** — symmetry-protected BIC or strongly under-coupled qBIC at $\omega_2 \approx 2\omega_1$. Aim for $Q_2 \gtrsim 4{,}000$; sensitivity in some applications scales as Q_2^2.
2. **Co-localisation** — both modes must share a meta-atom volume with high tensor overlap. Target $\eta_{\mathrm{geom}} \gtrsim 0.1$. Random pairings give $\eta \sim 10^{-3}$, which kills the cascade.
3. **Tensor channel** — for LiNbO$_3$ (point group $3m$): exploit d_{33} with field along the optic axis (z-cut, e-polarised modes). Confirm the chosen polarisation geometry gives non-vanishing $\int \chi^{(2)}_{ijk} f^*_{2,i} f_{1,j} f_{1,k}$.
4. $\Delta\omega$ **tuning mechanism** — electro-optic (Pockels, ~GHz/V), thermo-optic (~ 1.5 GHz/K), or strain. Need $\sim$few γ_2 of tuning range (~ 300 GHz at $Q_2 = 4{,}000$).
5. **FM port** — moderate $Q_1 \sim 10^2$–10^3 for efficient in-coupling. Critical / over-coupled at ω_1 depending on application (over-coupled for SA / limiter, critical for bistability).